Module 2 Statistical Evaluation Report for Topographic Variables

Visual-statistical interpretation of elevation, slope, and aspect layers

Scope	This report evaluates the elevation, slope, and aspect outputs produced at the end of Module 2 using histograms, boxplots, a polar histogram, raster views, and scatter plots.

1. Executive Summary

The generated plots show that the elevation, slope, and aspect variables contain meaningful information in terms of both distributional behavior and spatial pattern. Elevation is right-skewed and concentrated at mid-range values. Slope shows a mostly unimodal and balanced distribution, whereas aspect exhibits a non-uniform directional structure.

The scatter plots indicate that there is no strong linear dependence among the three variables. This suggests that the variables do not fully duplicate one another and may provide complementary information for shrub presence prediction.

2. Boxplot Analysis

The boxplots summarize central tendency, spread, and outlier behavior for the three variables. Elevation has a median near 550 m and displays more upper-tail outliers at higher elevations. Slope has a median of roughly 23° and most observations fall within moderate slope ranges. Classical boxplot interpretation is limited for aspect because the variable is circular, with 0° and 360° being adjacent directions.

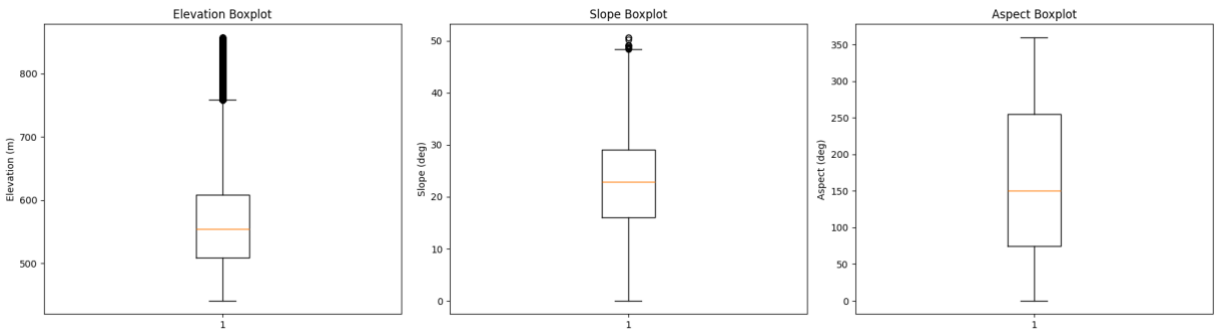

Figure 1. Boxplots of elevation, slope, and aspect.

3. Histograms and Directional Distribution

The elevation histogram is clearly right-skewed, indicating that mid-range elevations dominate the study area while very high elevations are less common. The slope histogram is more balanced and mostly unimodal. The aspect histogram is not uniform; some directions are more frequent whereas others are less represented.

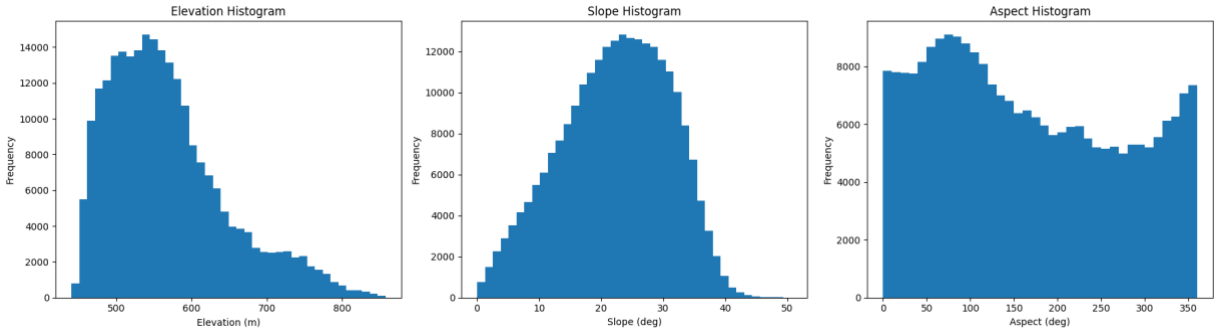

Figure 2. Histograms of elevation, slope, and aspect.

For aspect, the polar plot represents directional information more appropriately than linear axes. The plot suggests greater representation of roughly northeast-to-east directions, while some southwest-oriented sectors appear less frequent. This indicates that terrain orientation is not random across the study area.

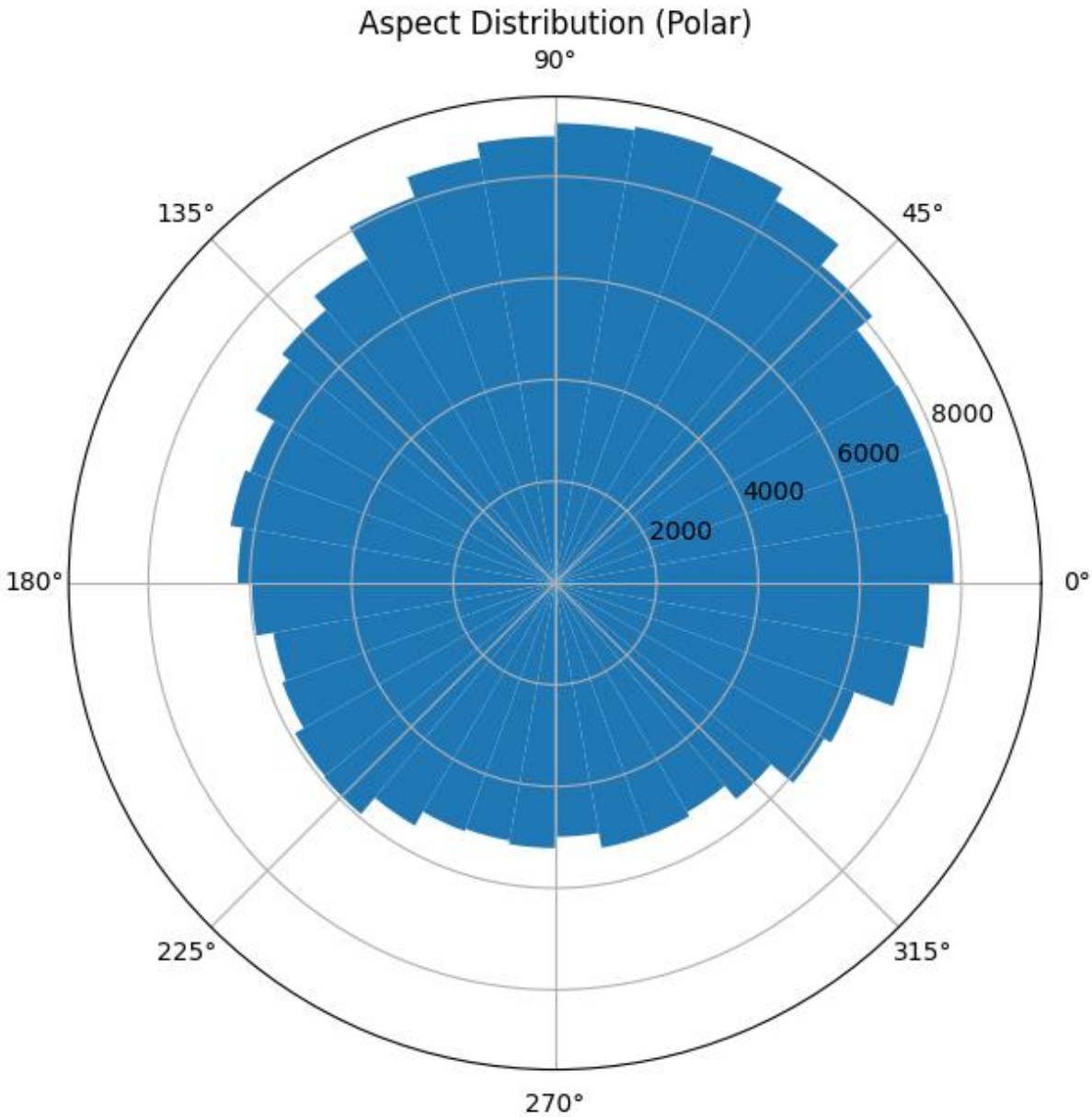

Figure 3. Polar distribution of aspect.

4. Spatial Patterns

The raster views reveal not only numerical distributions but also the spatial organization of the variables. The elevation layer shows a broad-scale elevation gradient. The slope layer clearly highlights drainage networks as well as hillslope and ridge transitions. The aspect layer appears more fragmented and banded because it is highly sensitive to small-scale changes in orientation.

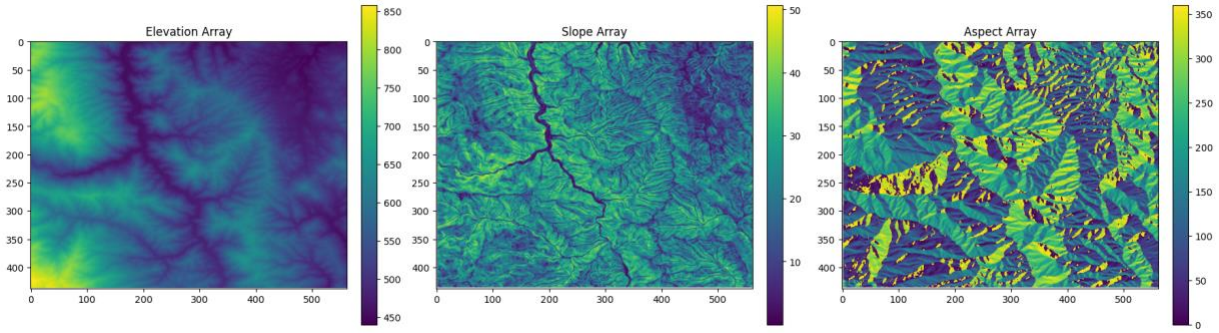

Figure 4. Raster views of elevation, slope, and aspect.

5. Relationships Among Variables

The scatter plots show a weak but not entirely random structure for the elevation–slope relationship. In contrast, no clear linear pattern is visible for elevation–aspect or slope–aspect. This suggests that the three variables retain a degree of complementarity as model inputs.

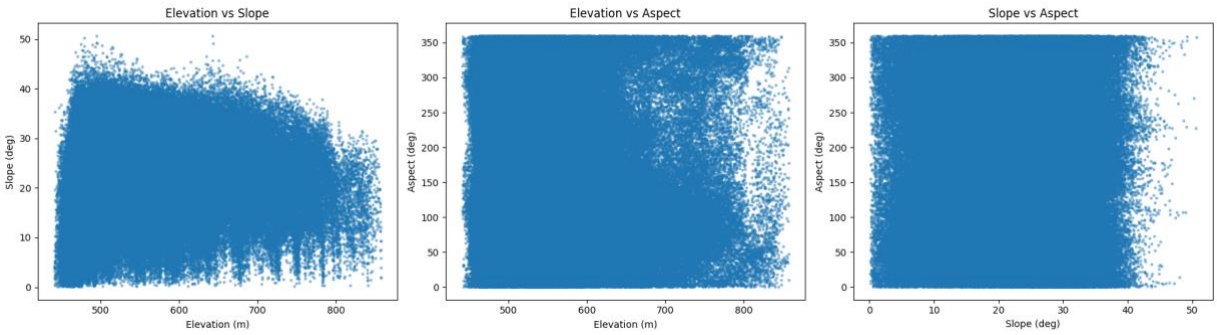

Figure 5. Scatter plots for variable pairs.

6. Implications for Shrub Presence Prediction

- Slope may be one of the most informative topographic variables because of its indirect links to surface stability, water retention, and erosion.
- Elevation is a useful second variable because it captures microclimatic and environmental gradients.
- Aspect is informative, but it is usually safer to model it as sin(aspect) and cos(aspect) rather than as raw values from 0 to 360.
- These three variables provide a suitable starting basis; however, for an ecological target such as shrub presence, additional variables may improve performance in later stages.

7. Additional Visual

The basemap overlay confirms that the generated aspect layer is spatially aligned with the study area and that the orientation information also forms meaningful patterns at the map level.

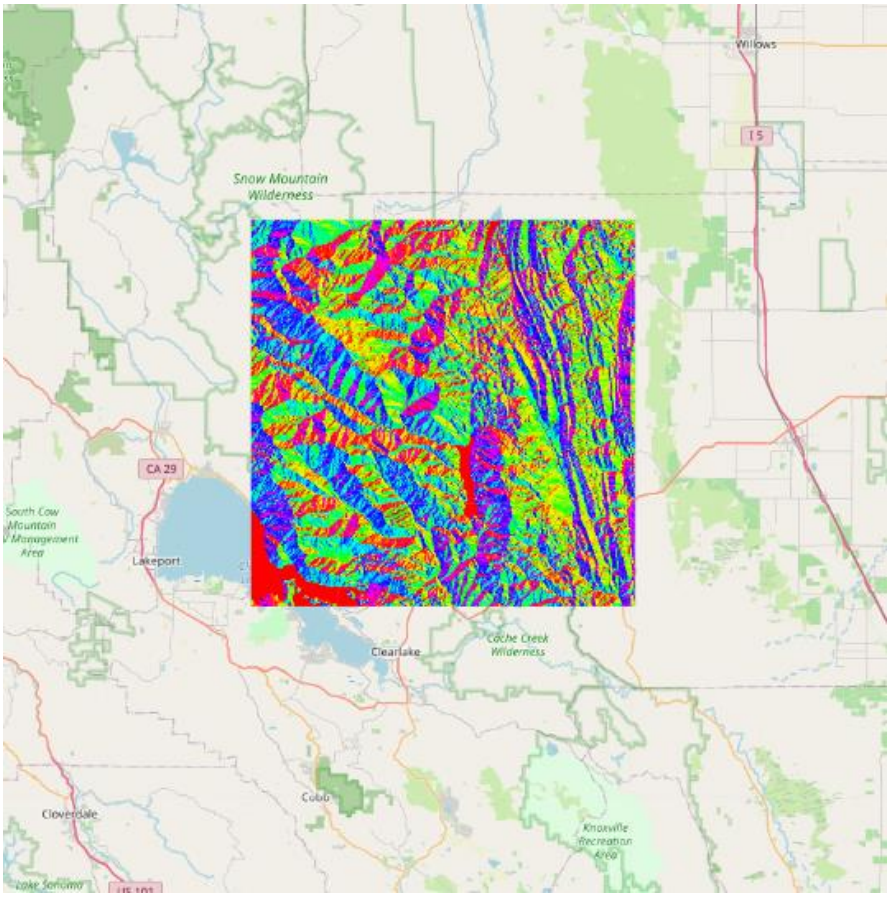

Figure 6. Aspect layer displayed over the basemap.

8. Conclusion

Overall, the shared plots indicate that the Module 2 outputs are consistent, interpretable, and usable for modeling. Slope appears to be the strongest topographic candidate, elevation provides strong supporting context, and aspect contributes additional information when encoded carefully. A sensible next step would be to test a baseline classifier using only these three variables and then introduce new variables only if the results require it.

Date prepared: 19 April 2026

NAIP-Based Vegetation and Shrub Analysis Report

1. Executive Summary

This report presents a detailed visual and statistical evaluation of NAIP imagery for vegetation analysis. The NDVI statistics (mean ≈ 0.25) indicate moderate vegetation density. The dataset contains strong vegetation signals, but shrub-specific separation requires additional features.

2. Spectral Band Histograms

The histograms reveal the distribution of spectral values. Red and Green bands are slightly right-skewed, indicating dominance of moderate reflectance values. The NIR band shows a wider range, confirming its sensitivity to vegetation reflectance.

3. NDVI Distribution

The NDVI histogram is concentrated in positive values (0.1–0.4), indicating that vegetation dominates the study area. Extreme values correspond to water (-) and dense vegetation (+).

4. Correlation Analysis

The correlation matrix shows strong redundancy among RGB bands (>0.9), while NIR provides more independent information. This highlights the importance of NIR for vegetation detection.

5. Spatial Visualization

RGB and false-color composites reveal clear spatial patterns. Vegetation appears strongly in the false-color image, while NDVI further enhances vegetation distribution.

6. Scatter Analysis (Red vs NIR)

The scatter plot shows a non-linear clustered relationship. Vegetation pixels occupy higher NIR values, while non-vegetation remains in lower ranges.

7. Implications for Shrub Detection

NDVI is effective for vegetation detection but insufficient for distinguishing shrubs from trees. Additional features such as texture or height (LiDAR) are recommended.

8. Conclusion

The dataset provides a strong basis for vegetation mapping. NIR and NDVI are the most informative features, while RGB contributes limited independent information.

Figures

Figure 1. Spectral band histograms.

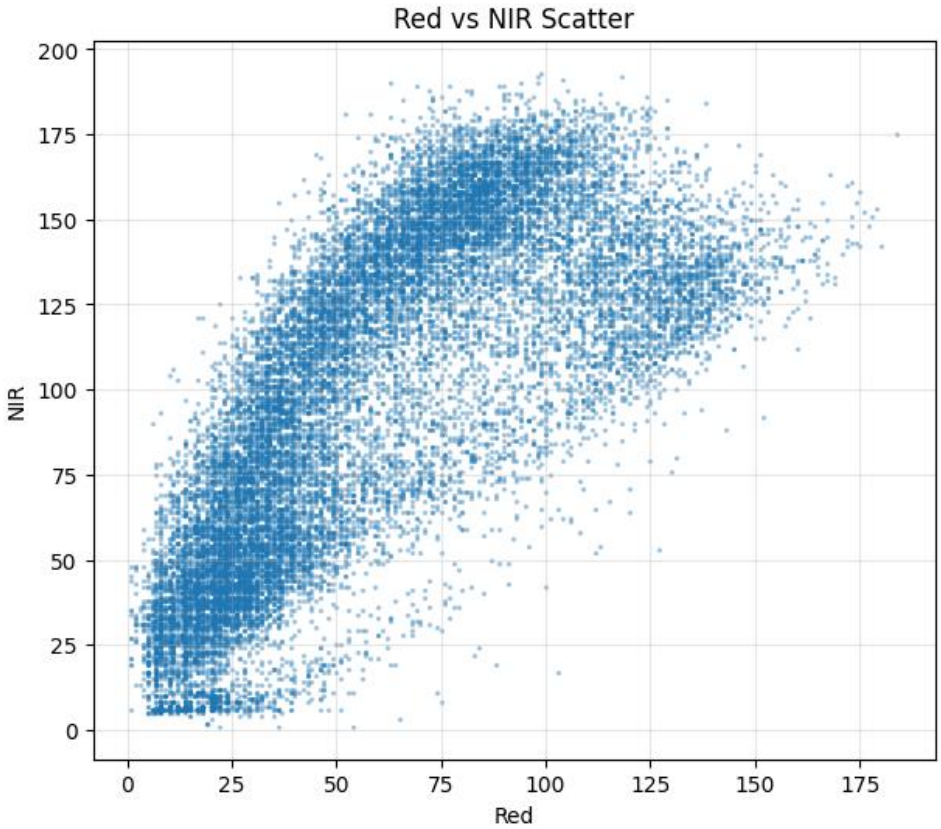

Figure 2. NDVI histogram.

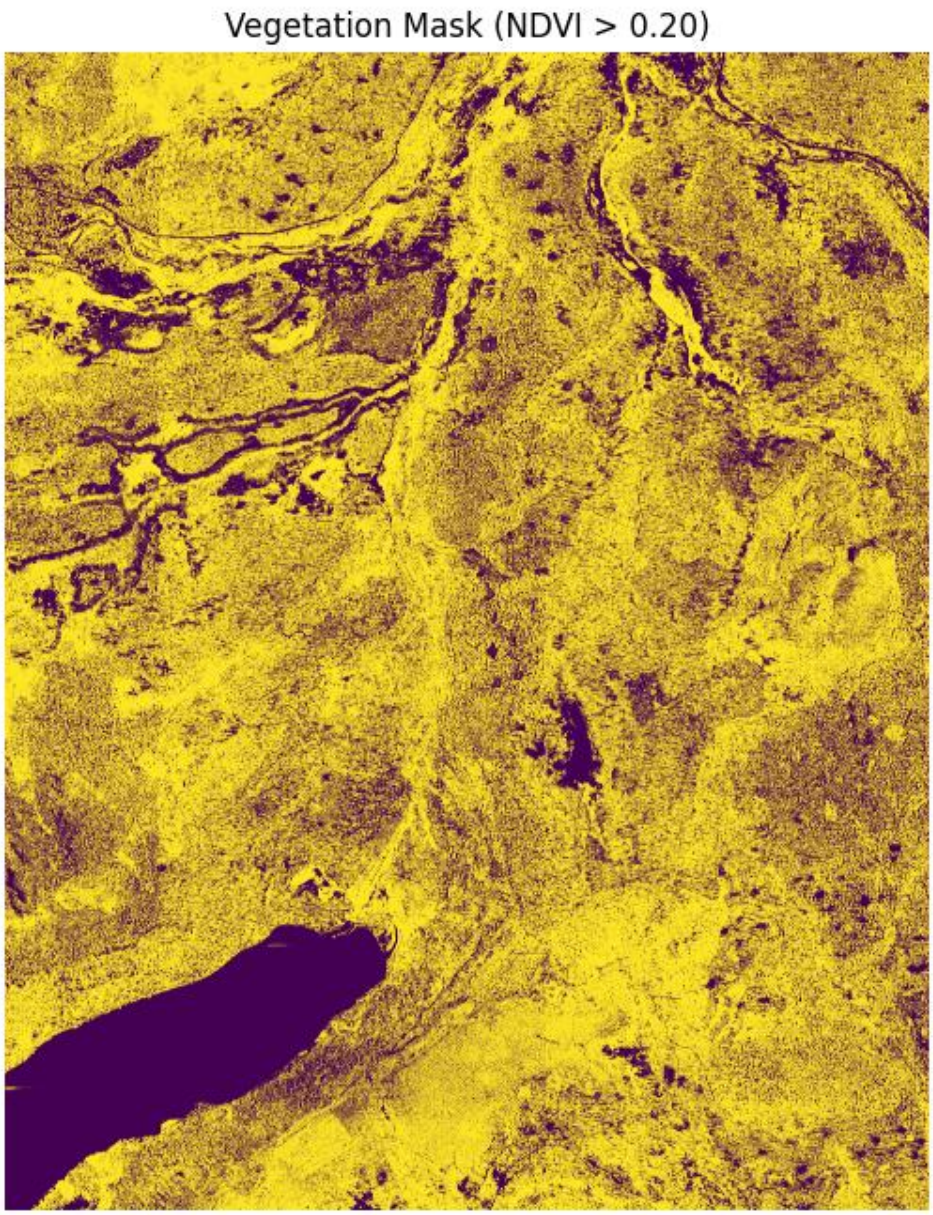

Figure 3. Correlation matrix.

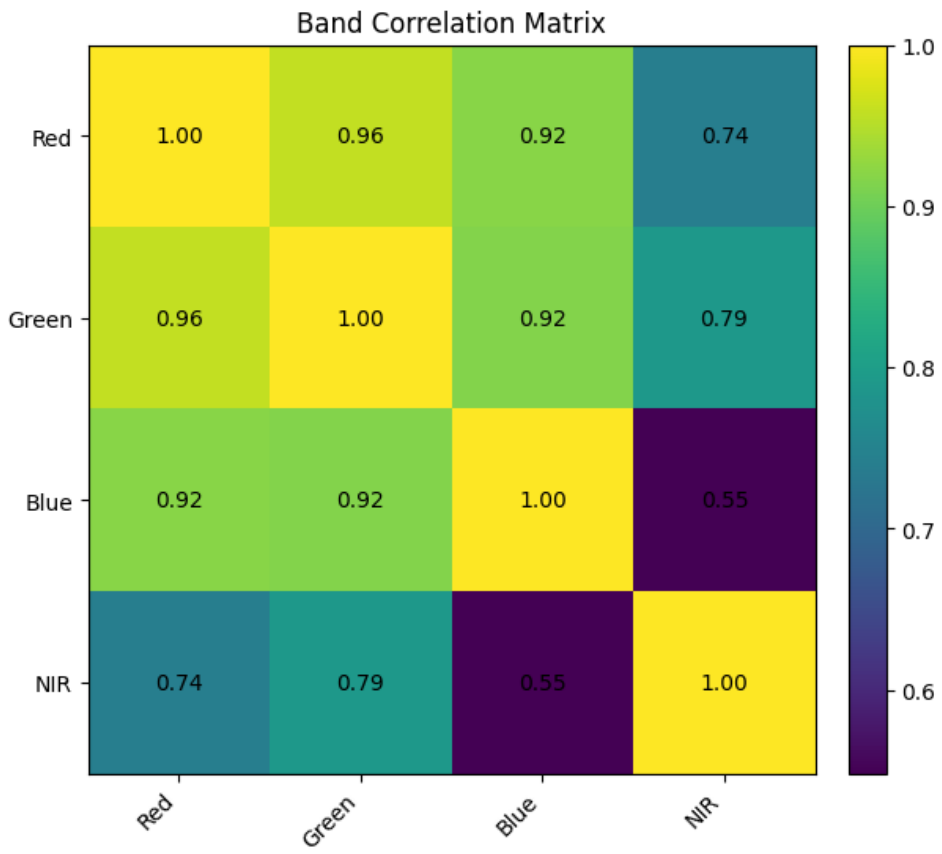

Figure 4. RGB composite.

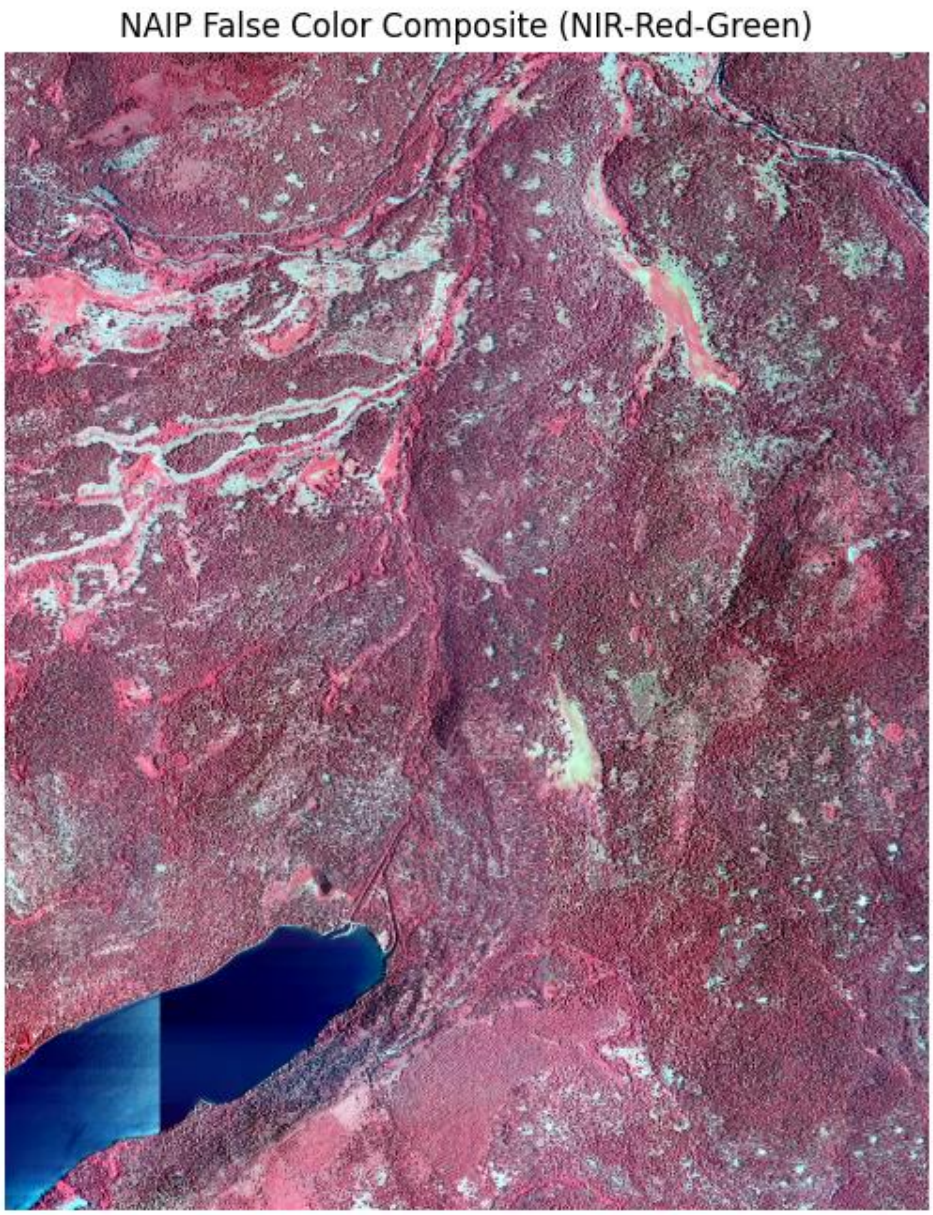

Figure 5. False color composite.

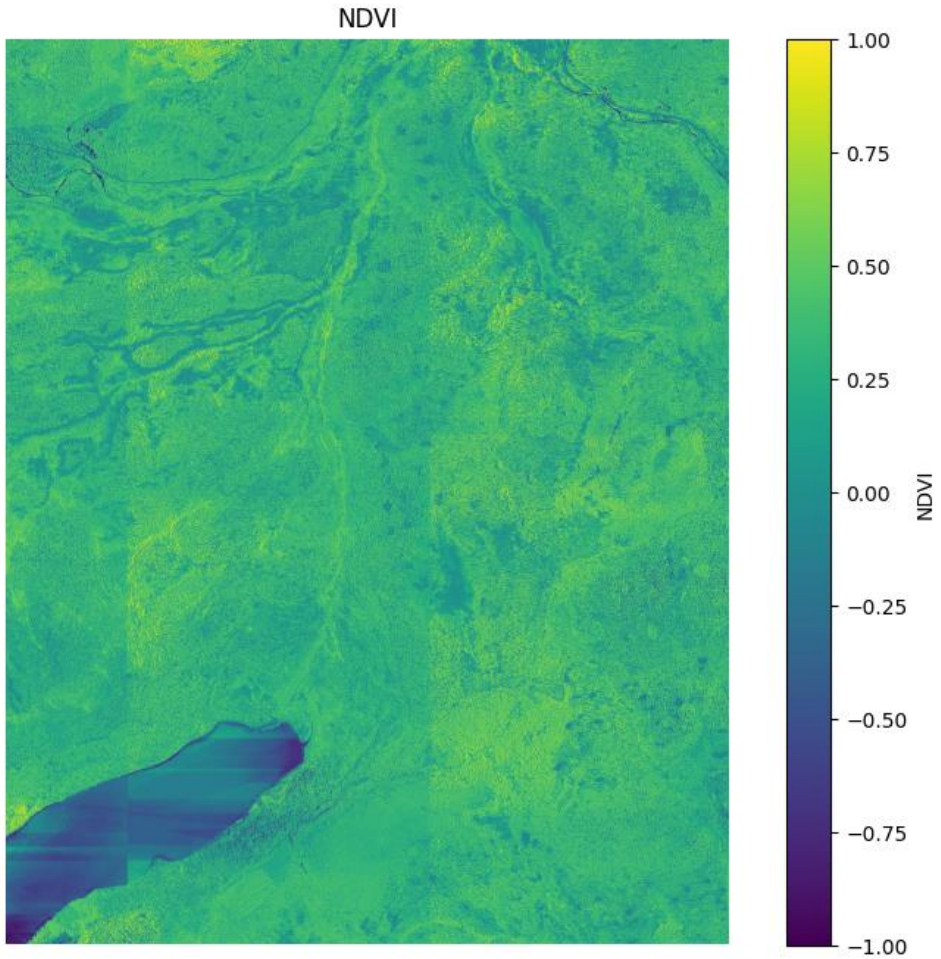

Figure 6. NDVI map.

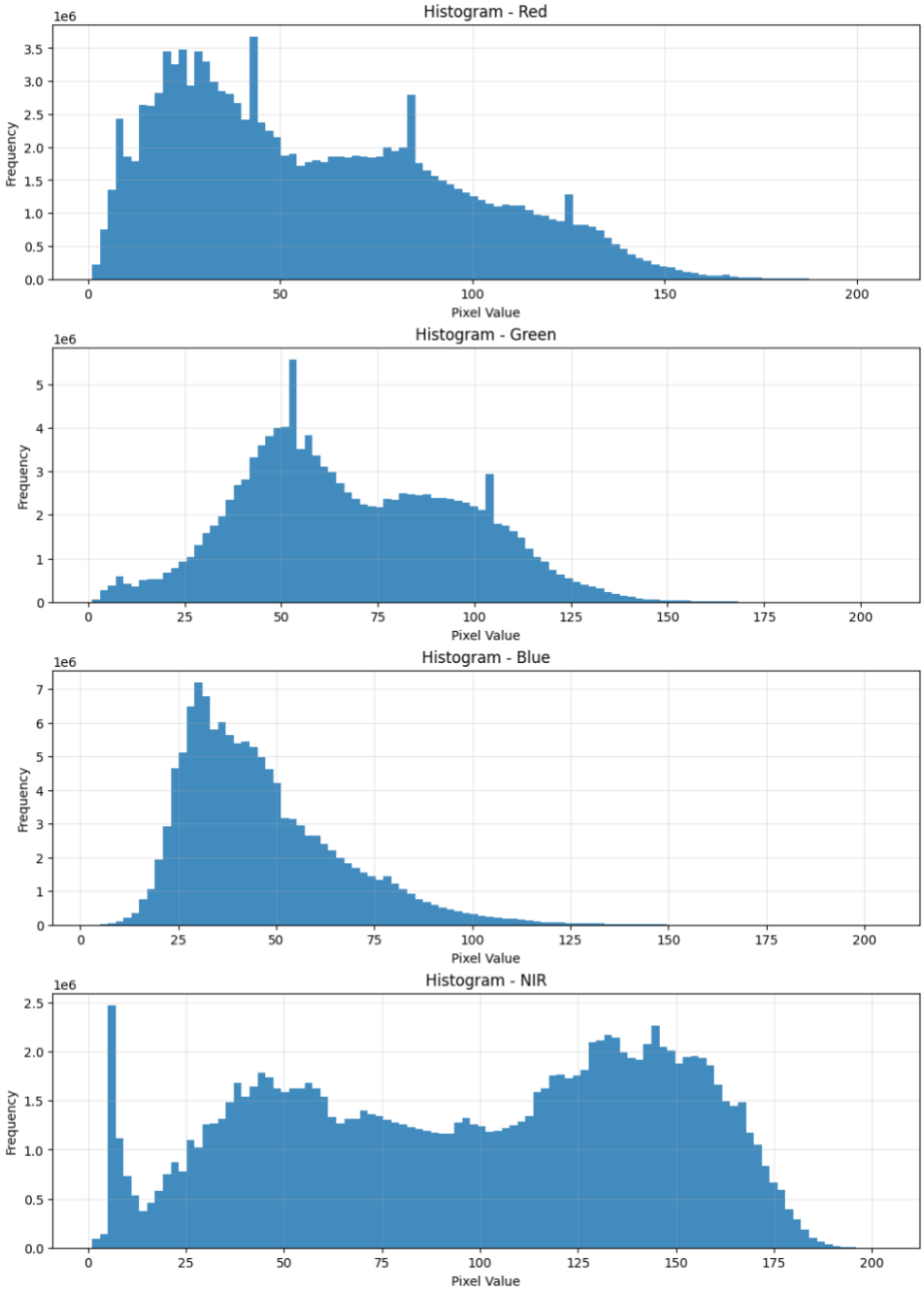

Figure 7. Red vs NIR scatter plot.